

All Parametric Survival Models

METIMMOX-1 Economic Evaluation

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Table of contents

1	Overview	1
2	Information Criteria	2
3	Coefficient Tables	4
4	Best-Fit Summary	5

1 Overview

This technical appendix lists every fitted parametric distribution for the single joint economic survival model.

Table 1: Fitted economic survival formulas

Endpoint	Formula
Overall survival	$\text{Surv}(\text{OSwk}, \text{Death}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} * \text{Rx} + \text{tmb_braf} * \text{Rx}$
Progression-free survival	$\text{Surv}(\text{PFSwk}, \text{Progression}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} * \text{Rx} + \text{tmb_braf} * \text{Rx}$

2 Information Criteria

Table 2: Information criteria for all fitted distributions

Distribution	AIC	BIC	Endpoint	Delta_AIC	Delta_BIC
weibull	666.98	686.96	Overall survival	0.00	0.00
weibullph	666.98	686.96	Overall survival	0.00	0.00
gamma	667.31	687.28	Overall survival	0.32	0.32
llogis	668.86	688.83	Overall survival	1.87	1.87
gengamma	668.97	691.17	Overall survival	1.99	4.21
lognormal	670.97	690.95	Overall survival	3.99	3.99
genf	670.98	695.39	Overall survival	3.99	8.43
gompertz	673.66	693.63	Overall survival	6.67	6.67
exponential	686.14	703.89	Overall survival	19.15	16.93
lognormal	489.86	509.84	Progression-free survival	0.00	0.00
gamma	491.48	511.46	Progression-free survival	1.62	1.62
gengamma	491.68	513.87	Progression-free survival	1.82	4.04
weibull	492.54	512.52	Progression-free survival	2.68	2.68
weibullph	492.54	512.52	Progression-free survival	2.68	2.68
llogis	492.85	512.82	Progression-free survival	2.99	2.99
genf	493.68	518.09	Progression-free survival	3.82	8.26
gompertz	496.55	516.52	Progression-free survival	6.69	6.69
exponential	501.16	518.92	Progression-free survival	11.30	9.08

3 Coefficient Tables

Table 3: Coefficient estimates for all fitted parametric models

Endpoint	Distribution	Parameter	Estimate	SE	95% CI
OS	exponential	rate	0.0057	0.0058	[0.0008, 0.0415]
OS	exponential	Age	0.0073	0.0149	[-0.0218, 0.0364]
OS	exponential	sex1	0.2141	0.2677	[-0.3105, 0.7387]
OS	exponential	RxExperimental arm	0.2869	0.3701	[-0.4385, 1.0123]
OS	exponential	crp1	-0.5340	0.5502	[-1.6123, 0.5443]
OS	exponential	tmb_brafl	-0.1355	0.3975	[-0.9145, 0.6436]
OS	exponential	RxExperimental arm:crp1	-0.1318	0.6794	[-1.4634, 1.1997]
OS	exponential	RxExperimental arm:tmb_brafl	-0.1209	0.5541	[-1.2069, 0.9651]
OS	weibull	shape	1.7474	0.1883	[1.4147, 2.1583]
OS	weibull	scale	158.0336	94.1040	[49.1910, 507.7067]
OS	weibull	Age	-0.0043	0.0088	[-0.0216, 0.0130]
OS	weibull	sex1	-0.1969	0.1556	[-0.5019, 0.1082]
OS	weibull	RxExperimental arm	-0.2573	0.2113	[-0.6715, 0.1569]
OS	weibull	crp1	0.4732	0.3132	[-0.1406, 1.0871]
OS	weibull	tmb_brafl	0.0589	0.2291	[-0.3901, 0.5078]
OS	weibull	RxExperimental arm:crp1	0.0566	0.3898	[-0.7074, 0.8206]
OS	weibull	RxExperimental arm:tmb_brafl	0.0729	0.3223	[-0.5587, 0.7046]
OS	weibullph	shape	1.7474	0.1883	[1.4147, 2.1583]
OS	weibullph	scale	0.0001	0.0002	[0.0000, 0.0023]
OS	weibullph	Age	0.0075	0.0154	[-0.0227, 0.0378]
OS	weibullph	sex1	0.3440	0.2744	[-0.1938, 0.8818]
OS	weibullph	RxExperimental arm	0.4496	0.3718	[-0.2791, 1.1783]
OS	weibullph	crp1	-0.8269	0.5524	[-1.9097, 0.2559]
OS	weibullph	tmb_brafl	-0.1029	0.4000	[-0.8868, 0.6810]
OS	weibullph	RxExperimental arm:crp1	-0.0989	0.6809	[-1.4333, 1.2356]
OS	weibullph	RxExperimental arm:tmb_brafl	-0.1275	0.5628	[-1.2305, 0.9756]
OS	llogis	shape	2.4310	0.2657	[1.9623, 3.0117]
OS	llogis	scale	92.4453	63.5361	[24.0363, 355.5507]
OS	llogis	Age	-0.0012	0.0099	[-0.0207, 0.0183]
OS	llogis	sex1	-0.1538	0.1777	[-0.5021, 0.1946]
OS	llogis	RxExperimental arm	-0.1625	0.2737	[-0.6989, 0.3739]
OS	llogis	crp1	0.2228	0.4011	[-0.5634, 1.0089]
OS	llogis	tmb_brafl	-0.0308	0.2912	[-0.6016, 0.5399]
OS	llogis	RxExperimental arm:crp1	0.3317	0.4766	[-0.6025, 1.2658]
OS	llogis	RxExperimental arm:tmb_brafl	0.2056	0.3730	[-0.5254, 0.9366]
OS	lognormal	meanlog	4.8783	0.6945	[3.5170, 6.2395]
OS	lognormal	sdlog	0.7411	0.0701	[0.6157, 0.8920]
OS	lognormal	Age	-0.0076	0.0098	[-0.0269, 0.0117]
OS	lognormal	sex1	-0.1874	0.1846	[-0.5491, 0.1743]
OS	lognormal	RxExperimental arm	-0.1423	0.2827	[-0.6965, 0.4118]
OS	lognormal	crp1	0.0737	0.3650	[-0.6418, 0.7891]
OS	lognormal	tmb_brafl	0.1219	0.2800	[-0.4268, 0.6707]
OS	lognormal	RxExperimental arm:crp1	0.5759	0.4598	[-0.3253, 1.4770]
OS	lognormal	RxExperimental arm:tmb_brafl	0.0858	0.3845	[-0.6677, 0.8393]
OS	gamma	shape	2.4303	0.4236	[1.7270, 3.4198]
OS	gamma	rate	0.0185	0.0123	[0.0051, 0.0678]
OS	gamma	Age	0.0036	0.0092	[-0.0145, 0.0217]
OS	gamma	sex1	0.1796	0.1675	[-0.1486, 0.5079]
OS	gamma	RxExperimental arm	0.2215	0.2365	[-0.2420, 0.6850]
OS	gamma	crp1	-0.3896	0.3306	[-1.0375, 0.2583]
OS	gamma	tmb_brafl	-0.0663	0.2474	[-0.5513, 0.4187]
OS	gamma	RxExperimental arm:crp1	-0.1711	0.4137	[-0.9820, 0.6397]
OS	gamma	RxExperimental arm:tmb_brafl	-0.0982	0.3451	[-0.7746, 0.5782]
OS	gompertz	shape	0.0100	0.0024	[0.0053, 0.0147]
OS	gompertz	rate	0.0022	0.0024	[0.0003, 0.0192]
OS	gompertz	Age	0.0095	0.0156	[-0.0210, 0.0401]
OS	gompertz	sex1	0.3436	0.2753	[-0.1960, 0.8832]
OS	gompertz	RxExperimental arm	0.4435	0.3739	[-0.2894, 1.1764]
OS	gompertz	crp1	-0.9710	0.5695	[-2.0872, 0.1452]
OS	gompertz	tmb_brafl	-0.1364	0.4005	[-0.9214, 0.6486]
OS	gompertz	RxExperimental arm:crp1	0.0537	0.6864	[-1.2916, 1.3990]
OS	gompertz	RxExperimental arm:tmb_brafl	-0.1193	0.5633	[-1.2235, 0.9848]
OS	gengamma	mu	4.9547	0.6275	[3.7249, 6.1846]
OS	gengamma	sigma	0.5897	0.1105	[0.4085, 0.8513]
OS	gengamma	Q	0.9047	0.5035	[-0.0822, 1.8916]
OS	gengamma	Age	-0.0032	0.0089	[-0.0206, 0.0142]
OS	gengamma	sex1	-0.1927	0.1615	[-0.5092, 0.1237]
OS	gengamma	RxExperimental arm	-0.2487	0.2217	[-0.6833, 0.1860]
OS	gengamma	crp1	0.4569	0.3345	[-0.1987, 1.1124]
OS	gengamma	tmb_brafl	0.0587	0.2336	[-0.3991, 0.5165]
OS	gengamma	RxExperimental arm:crp1	0.0772	0.4234	[-0.7526, 0.9069]
OS	gengamma	RxExperimental arm:tmb_brafl	0.0831	0.3298	[-0.5633, 0.7296]
OS	genf	mu	4.9592	0.6282	[3.7280, 6.1904]
OS	genf	sigma	0.5874	0.1114	[0.4050, 0.8510]

4 Best-Fit Summary

Table 4: Best-fitting parametric distributions by AIC

Endpoint	Distribution	AIC
Overall survival	weibull	666.98
Progression-free survival	lognormal	489.86

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